

Michael C. Puente

Research, design, and hands-on problem solving

B.S. Mechanical Engineering
The University of Texas at San Antonio
Expected Fall 2027

I'm grateful for the chance to pursue engineering through coursework, research, and practical work. These projects show what I have built, the questions I am investigating, and where I am continuing to learn.

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Understanding flow at Mach 6

UTSA | Numerical Evaluation of Surface Imperfections in Hypersonic Turbulent Boundary Layers

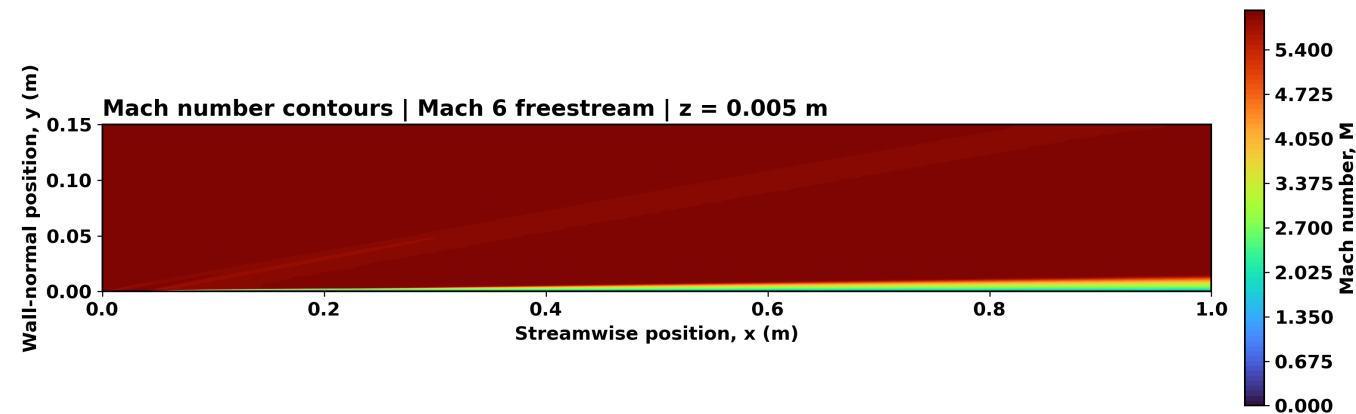
PURPOSE

I am developing a two-dimensional flat-plate baseline to examine boundary-layer growth, wall friction, and pressure changes. This foundation will guide my next objective: a rectangular-cavity simulation to study the effect of a surface imperfection.

MY WORKFLOW

OpenFOAM case setup and URANS simulation, ParaView visualization, and MATLAB post-processing. I use the outputs to connect near-wall flow behavior with the aerospace questions behind drag, loading, and heating.

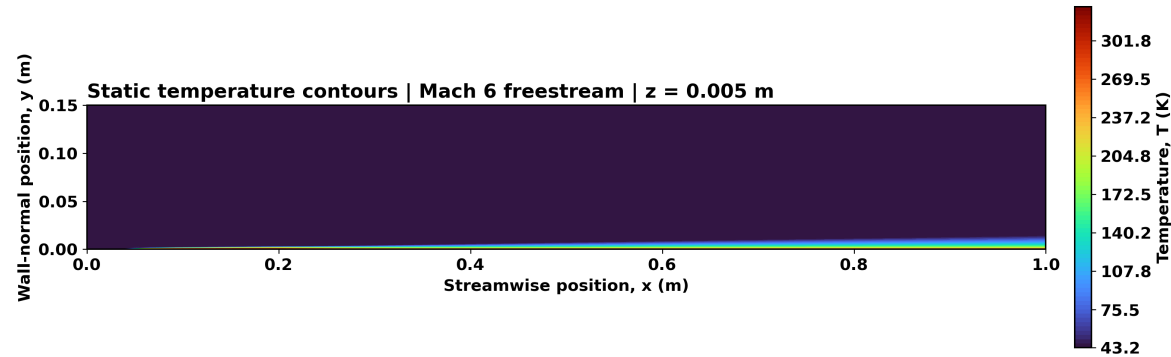
$M_\infty = 6$ | $U_\infty = 790.3 \text{ m/s}$ | $T_\infty = 43.18 \text{ K}$ | $p_\infty = 443.5 \text{ Pa}$ | $\rho_\infty = 0.0358 \text{ kg/m}^3$



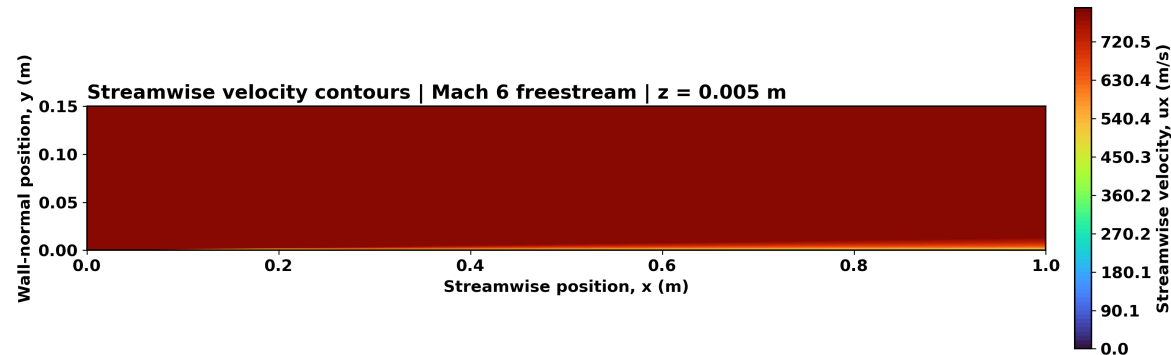
Local Mach number. The outer flow is near Mach 6. Lower Mach numbers near the plate show the developing boundary layer; the faint diagonal feature is consistent with an apparent shock.

Temperature and streamwise velocity

Flat plate | Flow left to right | x-y slice at $z = 0.005$ m | Preliminary simulation outputs



Static temperature, T (K). The outer flow is near 43.18 K, with a warmer near-wall region. This is gas temperature; it does not directly measure heat entering a vehicle surface.

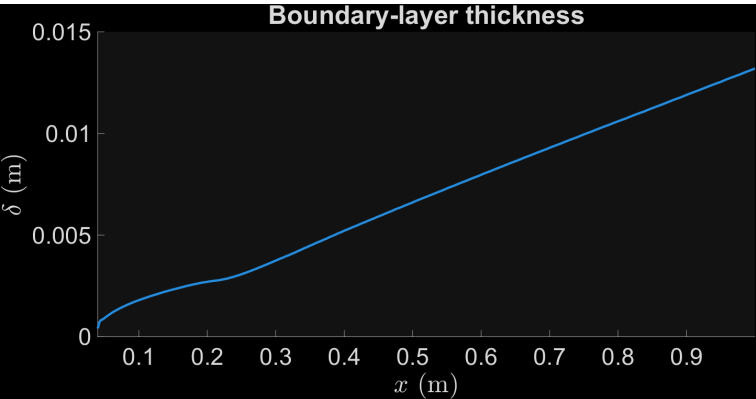


Streamwise velocity, u_x (m/s). Slower flow near the plate thickens downstream. The outer velocity is about 790.3 m/s: Mach 6 at $T_\infty = 43.18$ K, where ideal-air sound speed is approximately 131.7 m/s.

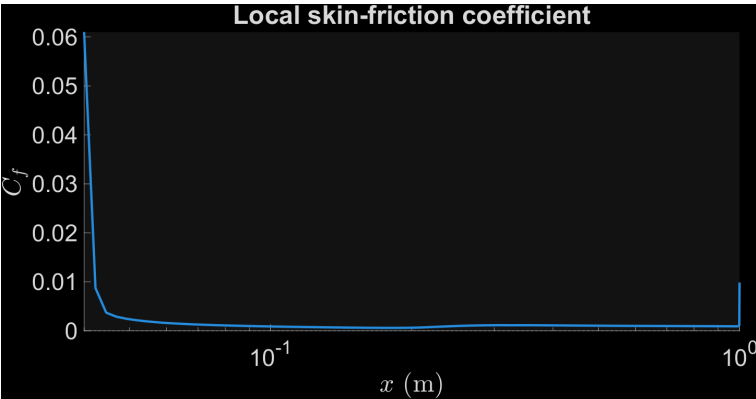
Mach contours use $|U|/\sqrt{\gamma RT}$, with $\gamma = 1.4$ and $R = 287$ J/(kg·K). Physical plot proportions are preserved.

From flow fields to engineering quantities

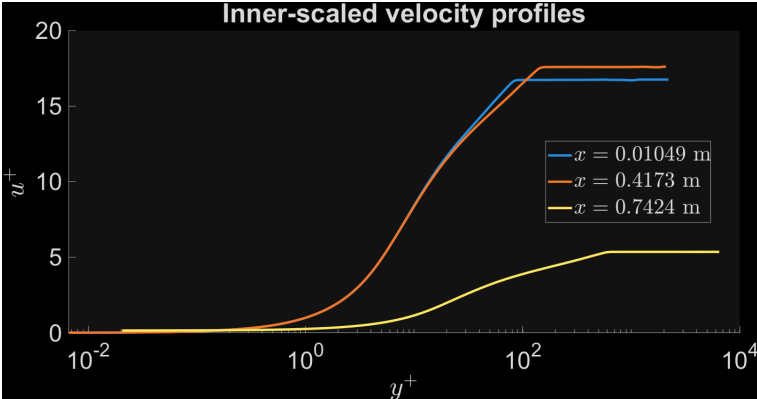
Boundary-layer development, skin friction, and velocity profiles



Boundary-layer thickness



Local skin friction



Velocity in wall units

AEROSPACE RELEVANCE

Near-wall flow influences skin-friction drag, surface pressure loads, and aerodynamic heating. Understanding how a surface imperfection changes that flow supports better questions about vehicle performance and thermal protection.

CURRENT PROGRESS AND NEXT OBJECTIVE

The flat-plate outputs show downstream boundary-layer growth. I am checking boundary values and numerical behavior before extending the study to a rectangular cavity and examining separation and recirculation.

Study status: Ongoing, with faculty guidance. Numerical verification and comparison with reference results remain part of the work. The adiabatic case assumes zero wall heat flux.

Contour processing retains the first record at repeated boundary coordinates; no smoothing is applied. Export time is not recorded in the CSV.

Two-Stage Spur-Gear Speed Reducer

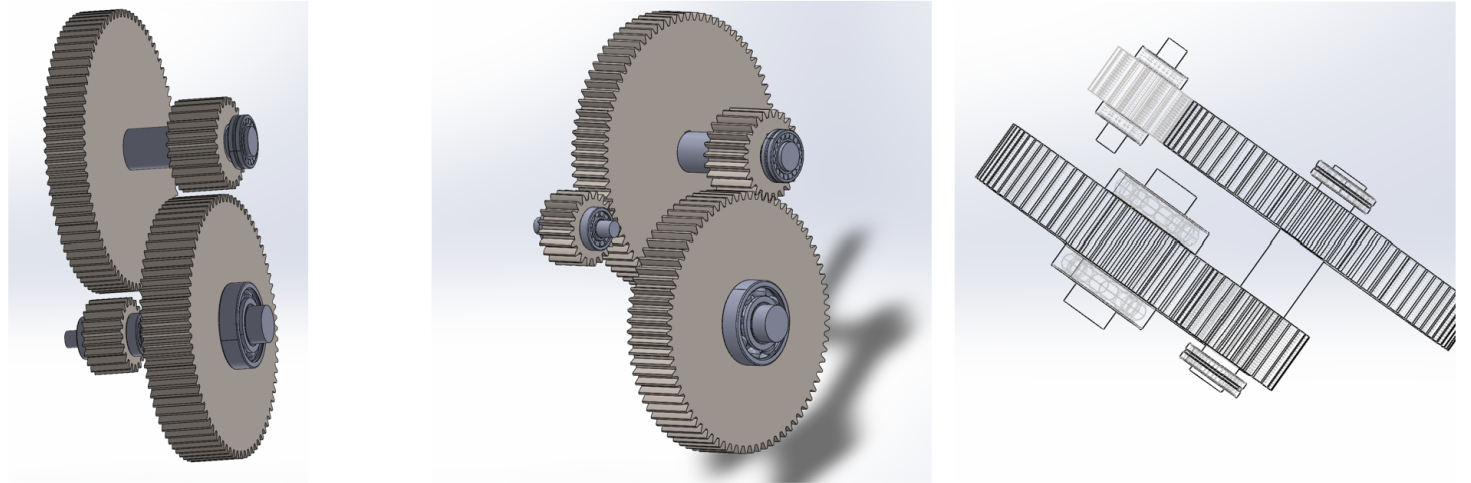
Machine Element Design | Completed independently | SolidWorks and MATLAB

DESIGN BRIEF

I independently designed an 82-hp reducer to step 1,200 rpm down to 100 rpm through 4:1 and 3:1 stages. I carried the requirement through gear sizing, shafts, bearings, keys, and the complete CAD assembly.

DESIGN APPROACH

- AGMA bending and contact-stress checks
- AISI 1045 cold-drawn steel shafts and ANSI keys
- SKF bearings selected from reaction loads and required dynamic ratings



SolidWorks assembly and section views

82 hp

INPUT POWER

12:1

OVERALL REDUCTION

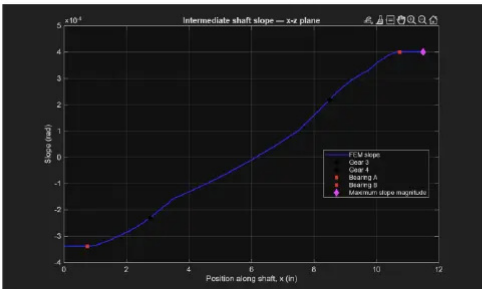
100 rpm

OUTPUT SPEED

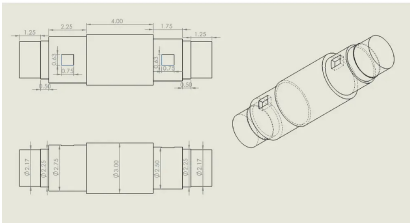
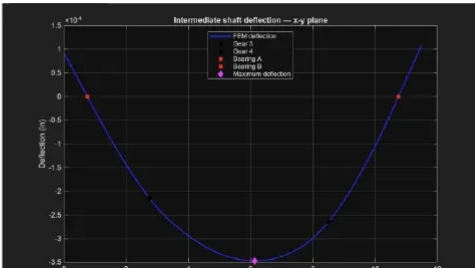
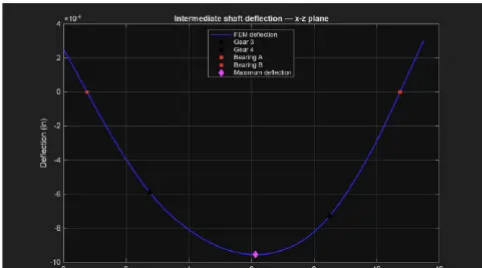
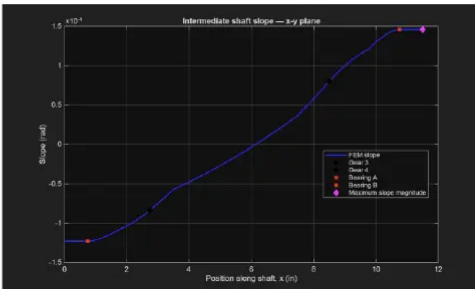
Checking the design against its requirements

Load analysis, fatigue, finite elements, and bearing selection

I.M Shaft x-z plane



I.M Shaft x-y plane



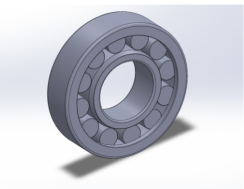
MATLAB beam-element FEM and dimensioned shaft geometry

Bearing Selection

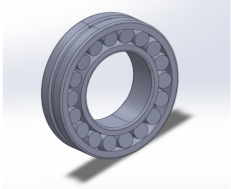
- Calculated bearing reaction forces from shaft loading analysis.
- Determined the required dynamic load rating (C_{10}) for each bearing.
- Selected standard SKF bearings with catalog ratings exceeding the required load.
- Verified all bearings satisfied the design life and reliability requirements.

Shaft	Bearing
Input	NJ307
Intermediate	22211E
Output	6315

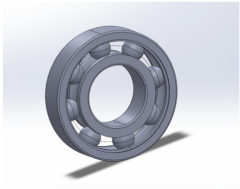
NJ307



22211E



6315



SKF NJ307 input, 22211E intermediate, and 6315 output bearings

CALCULATED DESIGN CHECKS

1.24 minimum gear-contact fatigue factor of safety

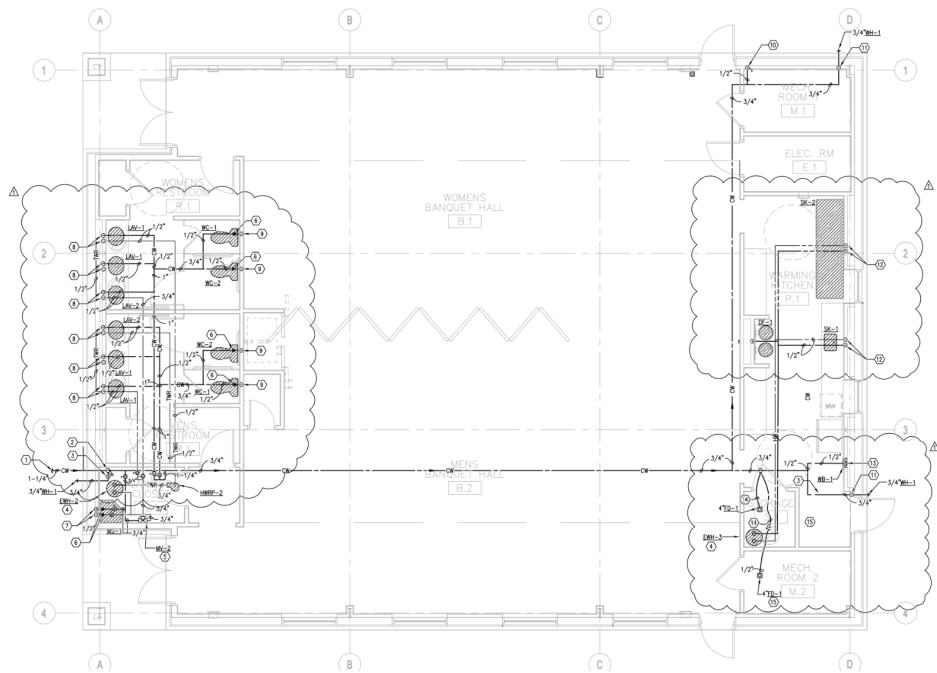
1.37 minimum shaft factor of safety

<0.001 in maximum intermediate-shaft deflection

<4.3 × 10⁻⁴ rad intermediate-shaft slope

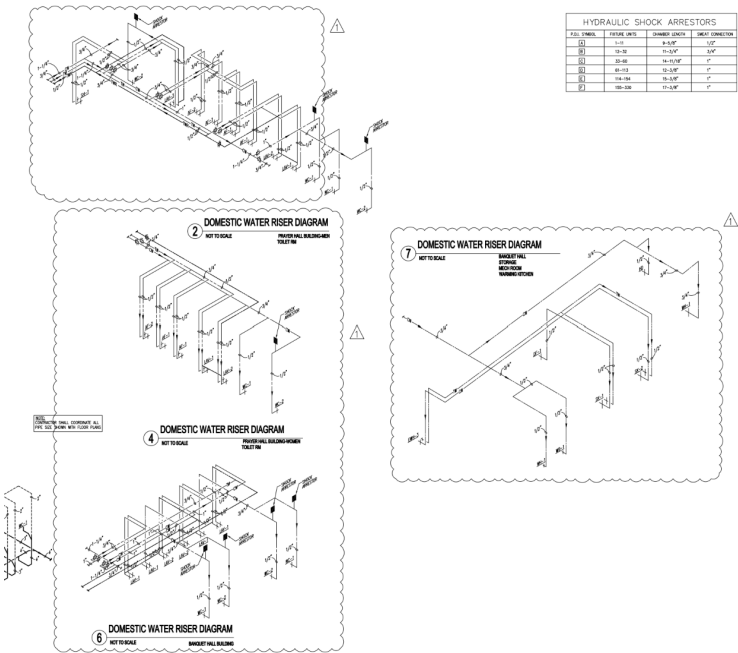
Masjid Al-Ekhlās permit revisions

My design and documentation within Delta 1 revision clouds | Permission granted for portfolio use



P.5 / Domestic-water plan

Design trigger. Architectural floor-plan changes required new and relocated fixtures and equipment, revised piping, and removal of obsolete work.



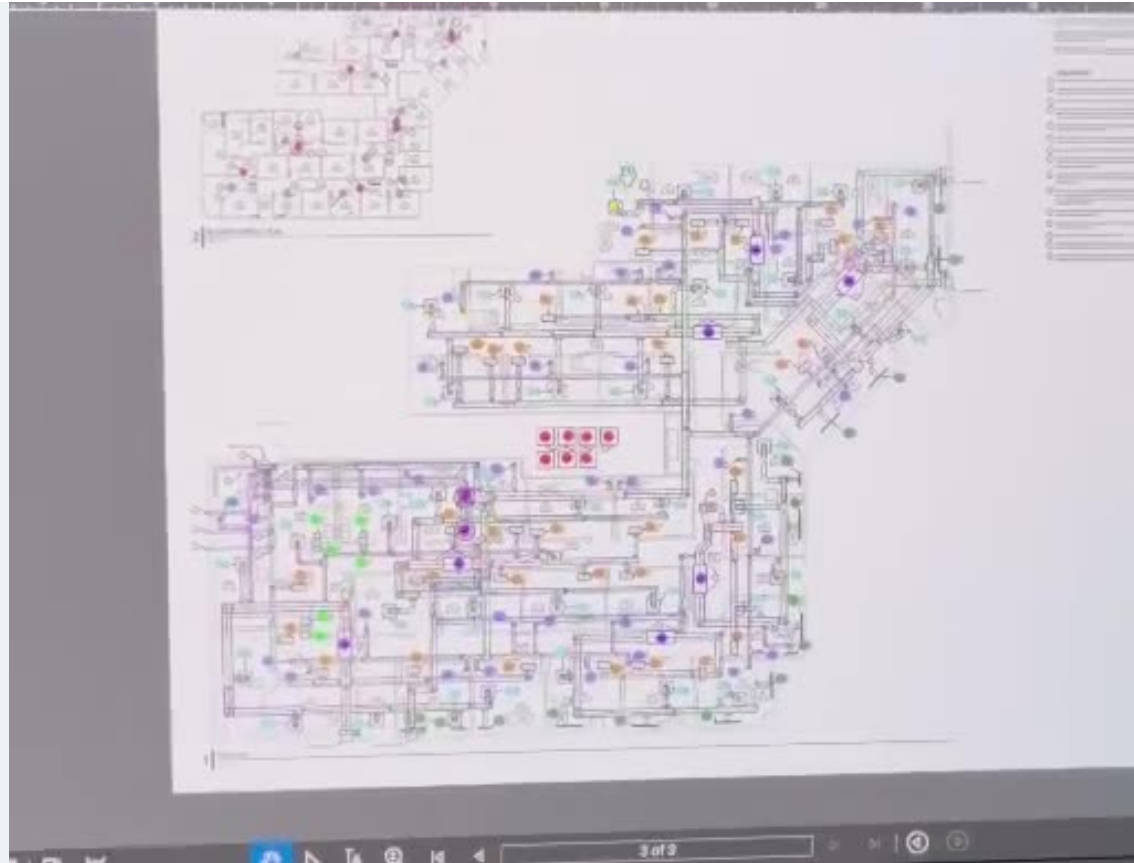
P.8 / Coordinated domestic-water risers

My contribution. Revised sanitary, vent, domestic-water and recirculation routing, then coordinated the plans with risers, details, schedules, and permit-response notes.

Attribution: Delta 1 revision-cloud sections are my work; the surrounding base drawings provide project context.

Commercial MEP review and coordination

Sur Texas LLC | Mechanical Engineering Intern | January 2025 - Present



Drawing-area excerpt from my mechanical markup walkthrough

ROUTINE MECHANICAL AND PLUMBING MARKUPS

- Review HVAC and plumbing bid drawings in Bluebeam Revu.
- Perform quantity takeoffs for ductwork, piping, equipment, fixtures, and accessories.
- Track revisions and scope changes to support estimating and coordination.

EQUIPMENT FAMILIARITY IN THE FIELD

Texas Air Products | Warehouse Associate

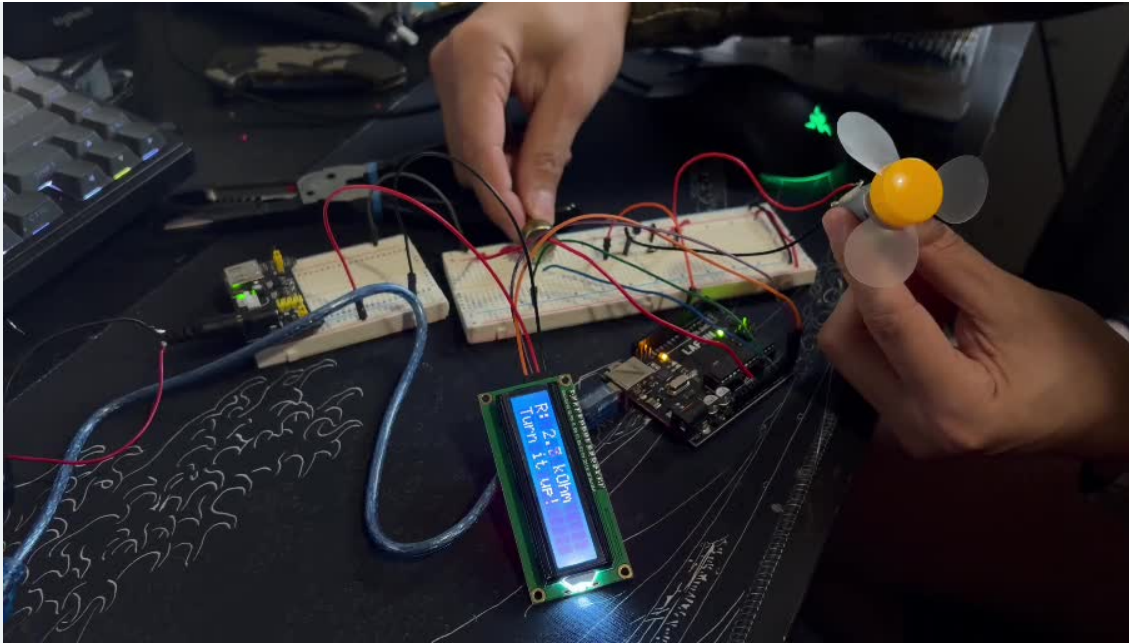
July 2026 - Present

Receive, identify, verify, stage, and load commercial HVAC products, including dampers, controls, air-distribution devices, VFDs/starters, curbs, and isolators.

[View the drawing walkthrough on my website](#)

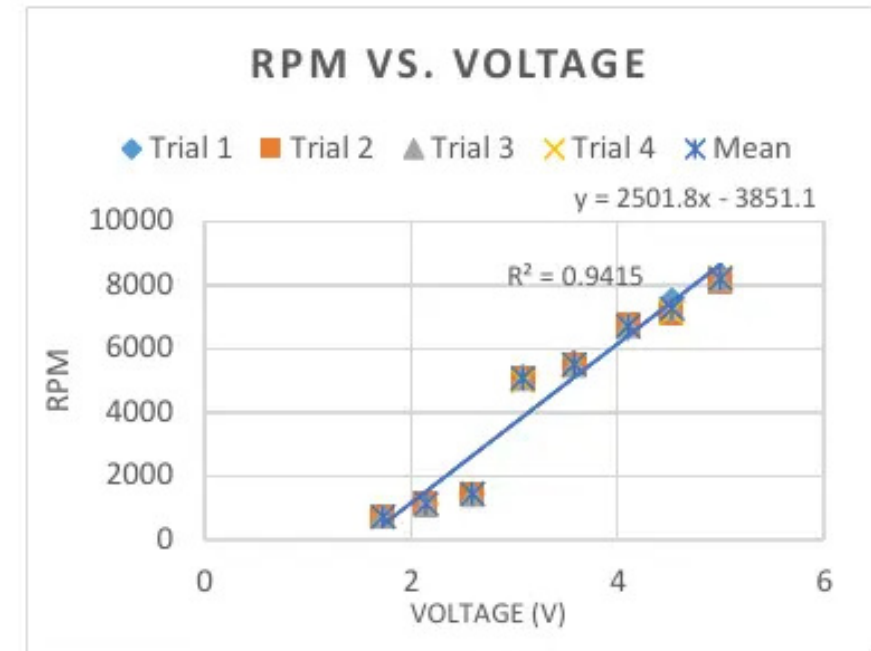
Arduino DC Motor Speed Control

Team laboratory project | PWM control, sensing, and experimental testing



Actual project setup / Video demonstration available online

Build. An Arduino Uno converts a 10 kΩ potentiometer input to PWM, driving the motor through a 2N2222 transistor and flyback diode, with an LCD displaying system values.

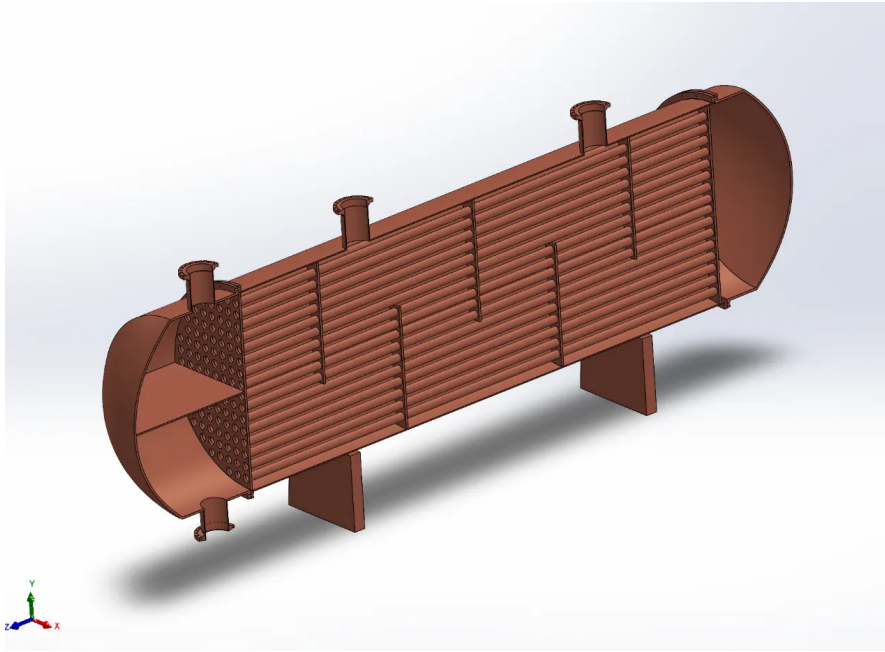


Experimental voltage-speed relationship

Test and result. Four trials per operating point yielded mean speeds of 743.6-8,225 rpm ($R^2 = 0.9415$). The observed low-voltage response prompted an L293D driver recommendation.

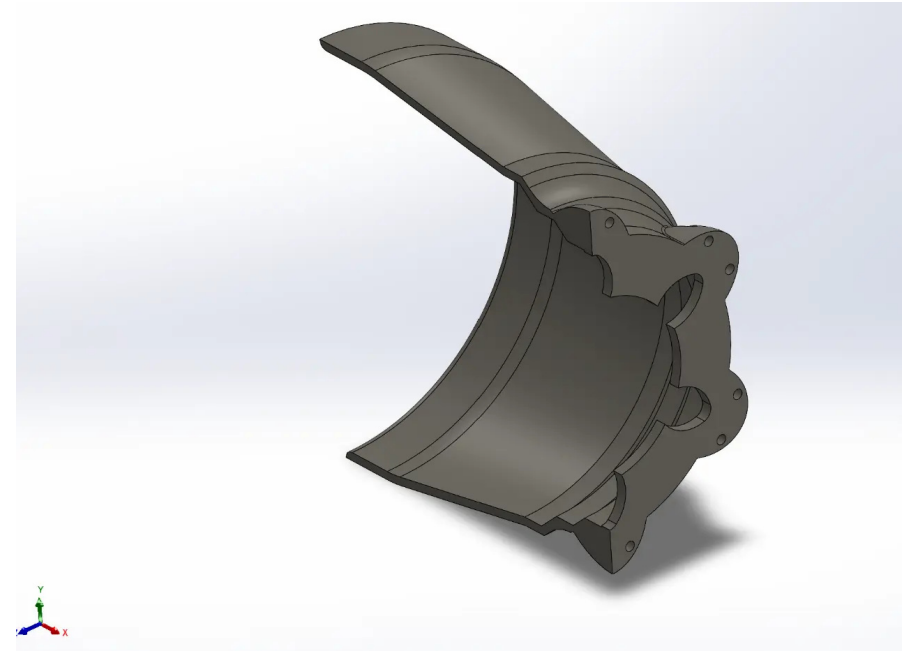
Learning through SolidWorks modeling

Assembly troubleshooting and single-component design practice



Shell-and-Tube Heat Exchanger

Tutorial-guided assembly. I identified a flow-separation issue in the tutorial configuration and independently revised the model so the shell-side and tube-side flow paths remained distinct.



Jet Engine Rear Casing

Work in progress: the rear casing only. I practiced revolved and swept geometry, cuts, fillets, circular patterns, and face edits. This is a CAD exercise, with no completed engine or performance claim.

Smokey | 1994 Ford F-150 restoration

Personal project | Fuel, starting, ignition, and electrical troubleshooting



Disassembly



Work in progress



Back together

WORK COMPLETED

Replaced both fuel tanks and pumps, injectors, throttle body, spark plugs, MAP sensor, battery, starter, starter solenoid, and power-steering reservoir. Cleaned the fuel rail and intake manifold and soldered injector-connection wiring.

WHAT I AM LEARNING

With family help and hands-on practice, I returned a non-running truck to operation. The project continues to teach me careful diagnosis, persistence, and attention to the details behind each repair.